

Architecture Decision Record (ADR): Model Selection for Trip Duration Prediction

Context

The objective of this project is to predict NYC Taxi Trip Durations. We have explored a dataset, performed feature engineering, and trained multiple regression models. This ADR documents the decision-making process for selecting the primary model to be deployed.

Decision

The **Linear Regression model** is selected as the primary model for deployment due to its superior R-squared performance compared to the initial Gradient Boosting Regressor, its simplicity, interpretability, and faster training/prediction times.

Why this model?

- **Performance:** The initial Linear Regression model achieved an R-squared of 0.03 and an MSE of 9,557,237.30 on the test set. While these metrics indicate low overall predictive power for *absolute* trip duration, it performed better than the initial Gradient Boosting Regressor which resulted in a negative R-squared of -0.12 and a higher MSE of 11,025,336.11.
- **Simplicity and Interpretability:** Linear Regression is a simpler model, making it easier to understand and debug. Its coefficients provide direct insights into the relationship between features and the target variable.
- **Computational Efficiency:** Linear Regression models are generally less computationally intensive to train and run predictions, which is beneficial for deployment in a production environment, especially if real-time predictions are required.
- **Baseline Performance:** Given the low R-squared values for both initial models, Linear Regression serves as a good, robust baseline that can be quickly deployed and iterated upon. The initial Gradient Boosting model, without tuning, performed worse, suggesting that its complexity was not immediately beneficial.
- **Monitoring and Retraining:** The Linear Regression model was successfully monitored for data drift, and a retraining mechanism was demonstrated to restore its performance after a simulated drift event.

What alternatives were considered?

- **Gradient Boosting Regressor (Untuned):** This model was implemented as an alternative. However, its initial performance (R-squared: -0.12) was significantly worse than Linear Regression. This suggests that without proper hyperparameter tuning, the added complexity of boosting algorithms does not necessarily lead to better results.
- **Gradient Boosting Regressor (Tuned - GridSearchCV):** Hyperparameter tuning was initiated for the Gradient Boosting Regressor using GridSearchCV. While the full execution was commented out in the provided notebook for brevity/computational reasons, the intent was to find optimal hyperparameters to improve its performance. In a real-world scenario, a properly tuned Gradient Boosting model would likely outperform Linear Regression. However, for this initial deployment, the untuned version's poor performance and the computational cost of extensive tuning led to prioritizing the simpler model first.

What trade-offs were accepted?

- **Lower Predictive Power:** By choosing Linear Regression with a low R-squared (0.03), we accept that the model has limited explanatory power for the variance in taxi trip duration. It serves as a baseline, and significant improvements will likely require more complex models (e.g., a well-tuned Gradient Boosting model, neural networks) and potentially more advanced feature engineering.
- **Simplicity over Potential Complexity Benefits:** We prioritize the simplicity, speed, and interpretability of Linear Regression over the potential for higher accuracy that a complex, well-tuned Gradient Boosting model might offer. This reduces initial development and deployment complexity.
- **No Immediate Hyperparameter Tuning:** The decision to deploy Linear Regression means we are not immediately investing in extensive hyperparameter tuning for more complex models, which would require more computational resources and time. This is a trade-off for faster initial deployment.
- **Sensitivity to Data Drift:** While a monitoring mechanism was implemented, Linear Regression can be sensitive to data distribution shifts. The documented data drift simulation demonstrated a significant performance drop, necessitating a retraining mechanism to mitigate this.

Consequences

- **Positive:** Quick deployment of a functional baseline model. Easier to monitor and debug. Provides a clear starting point for future model improvements.
- **Negative:** The deployed model may not provide highly accurate predictions for all scenarios, especially complex ones not captured by linear relationships. There is a clear path for future enhancement by incorporating a tuned, more powerful model.